

Conditional probability



1

a 0.23

b 0.73

2

a $x = 0$

b $x = 0.2, 0.3$

3

a

	2	4	5
3	5	7	8
7	9	11	12

b $\frac{2}{5}$

c 1

4

a

	1	2	4	6
1	1	2	4	6
2	2	4	8	12
3	3	6	12	18
4	4	8	16	24
5	5	10	20	30
6	6	12	24	36

b $\frac{1}{2}$

c $\frac{3}{10}$

d $\frac{1}{2}$

Dependent and independent events

5

a Independent

b Dependent

c Dependent

d Dependent

6

a Yes

b No

7 0.2

8

a 0.7

b Yes

c 0.5

9 $\frac{8}{33}$



10 a $\frac{32}{663}$

b $\frac{4}{39}$

11 a i $\frac{33}{16\,660}$

ii $\frac{1}{649\,740}$

b 2 598 960

12 a No

b $\frac{1}{6}$

13 a $\frac{1}{45}$

b $\frac{1}{44}$

c No

d Dependent

14 a $\frac{1}{51}$

b $\frac{1}{50}$

c Dependent

15 a $\frac{5}{18}$

b $\frac{1}{3}$

16 a $\frac{1}{64}$

b $\frac{11}{850}$

17 a $\frac{1}{3}$

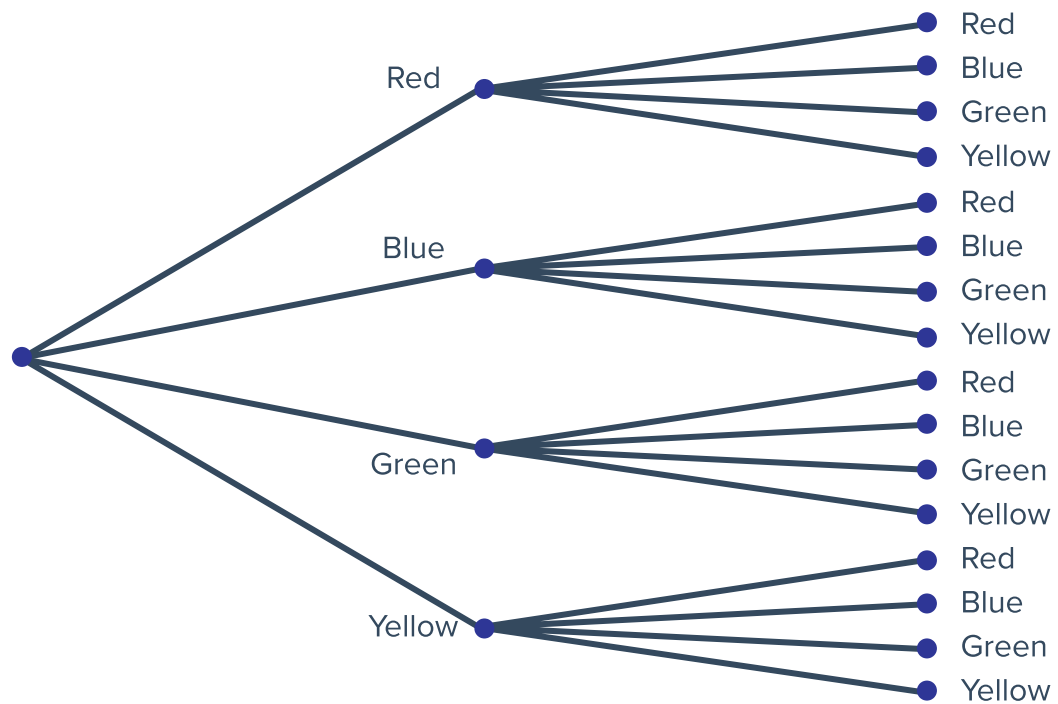
b $\frac{1}{12}$



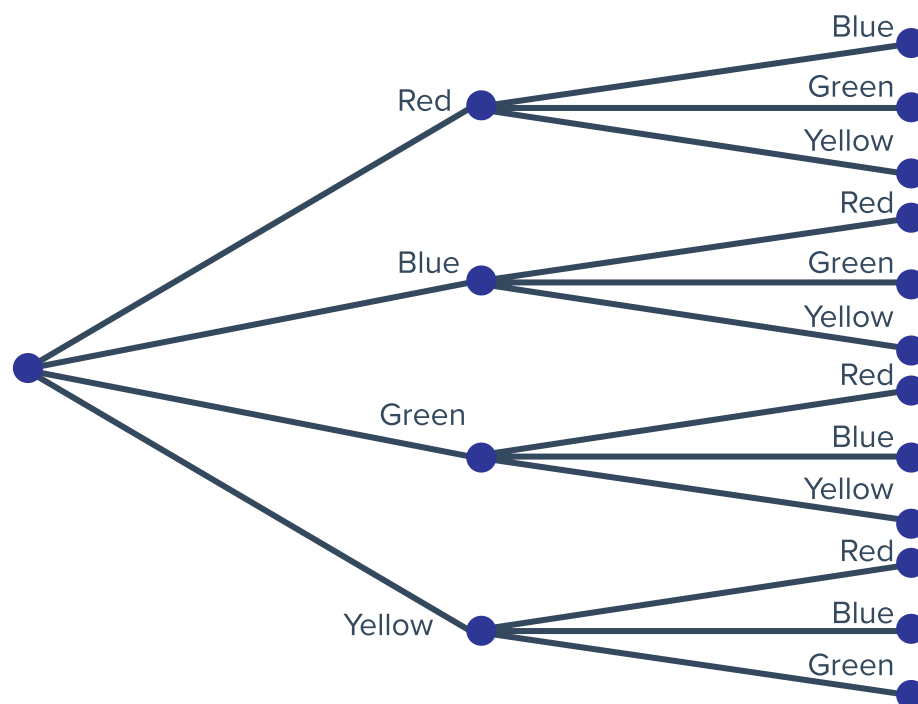
Tree diagrams and probability trees

18

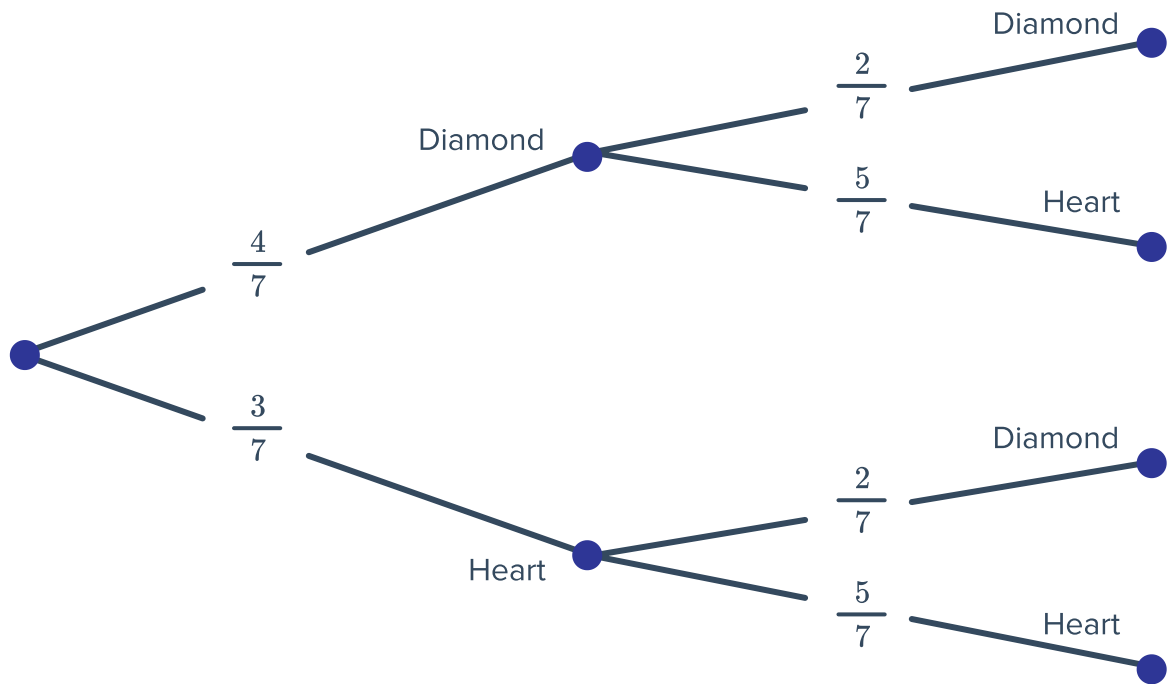
a



b

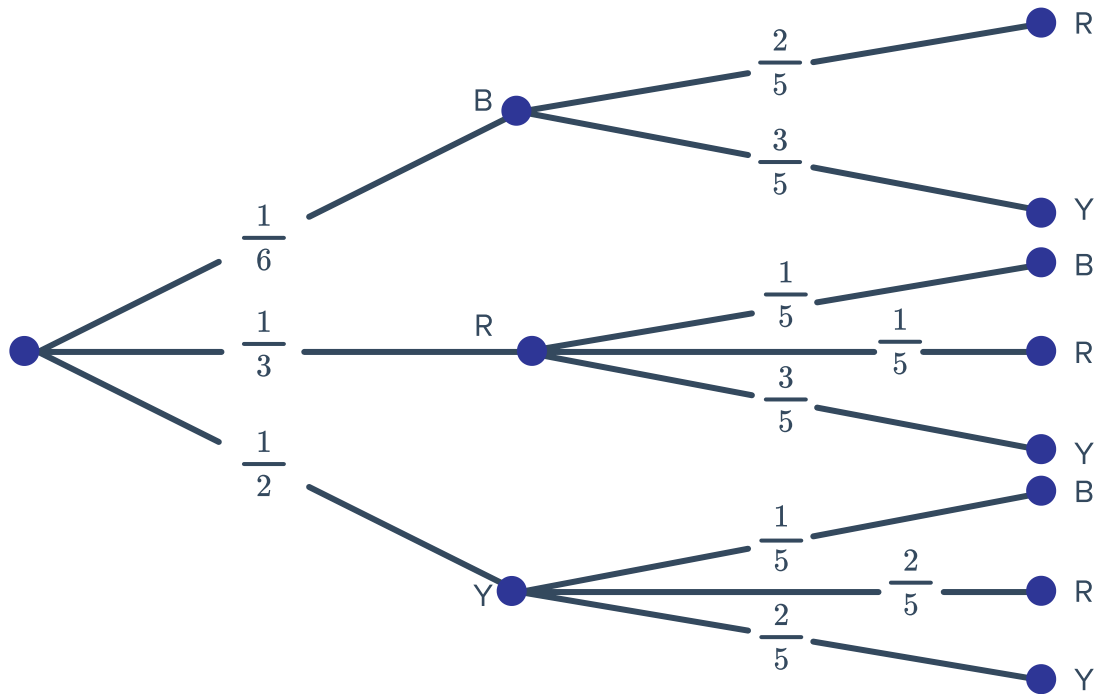


a



b $\frac{15}{49}$

a



b

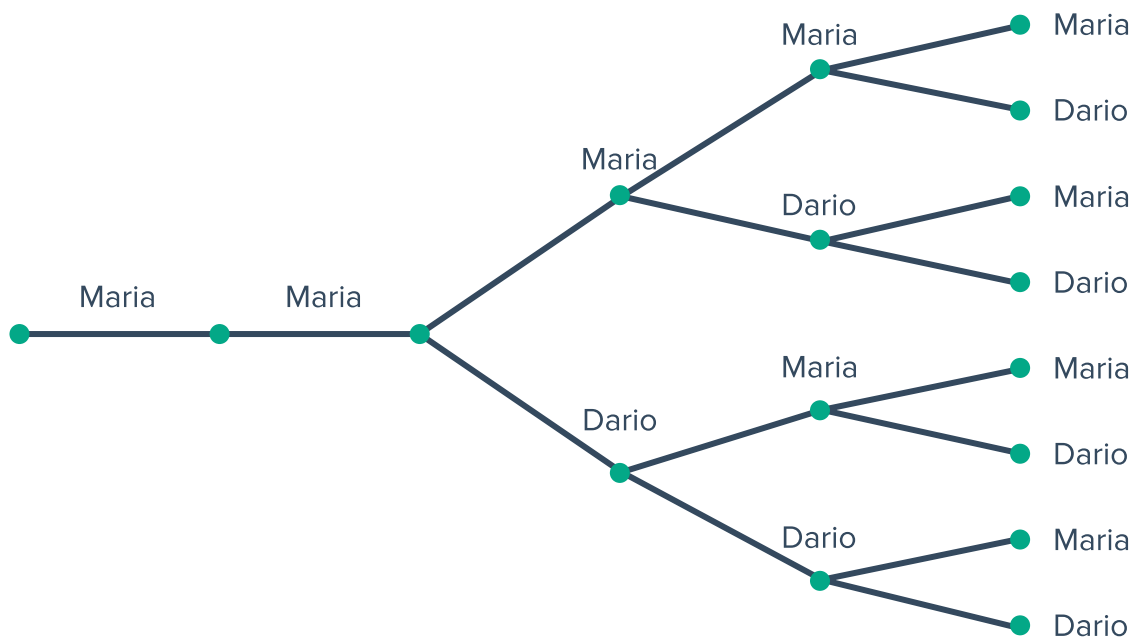
i $\frac{1}{10}$
v 0

ii $\frac{1}{15}$
vi $\frac{1}{5}$

iii $\frac{1}{15}$
vii $\frac{2}{5}$

iv $\frac{1}{5}$

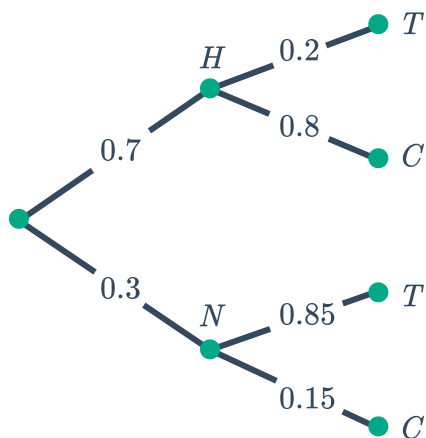
21 a



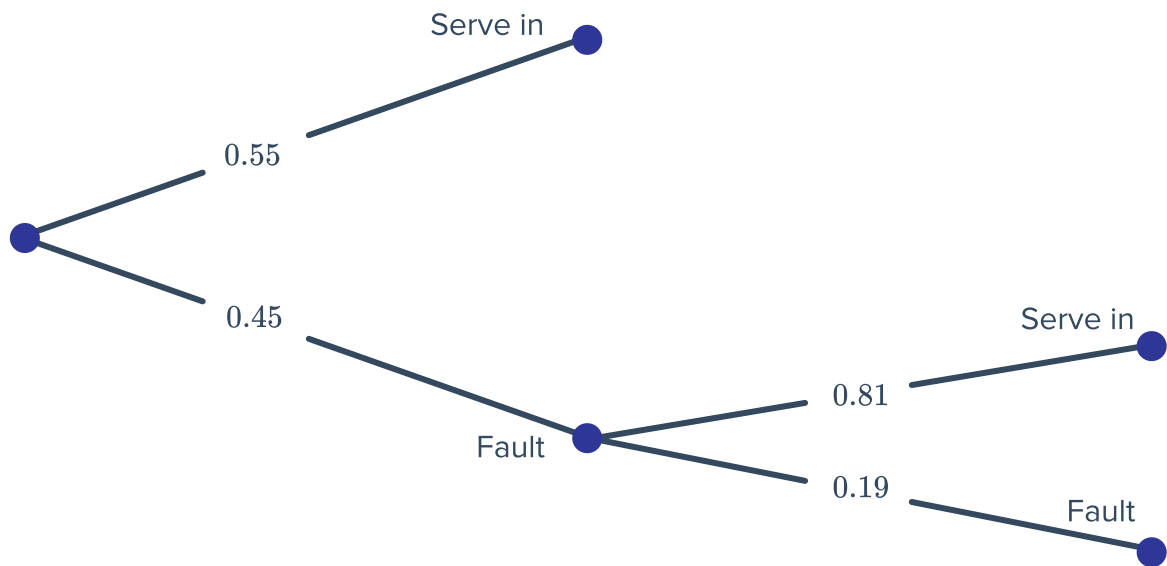
- b $\frac{2}{4}$
c $\frac{1}{4}$

22 a

b 0.35

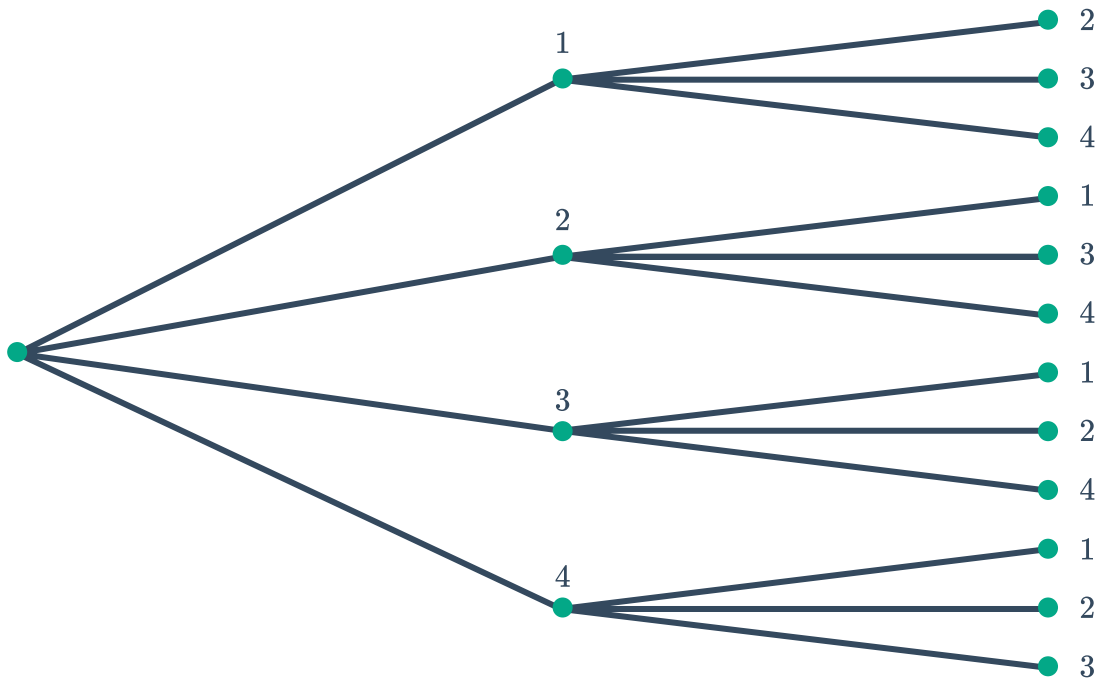


23
a



b 0.45

c 0.0855



b

i $\frac{1}{2}$
v 0

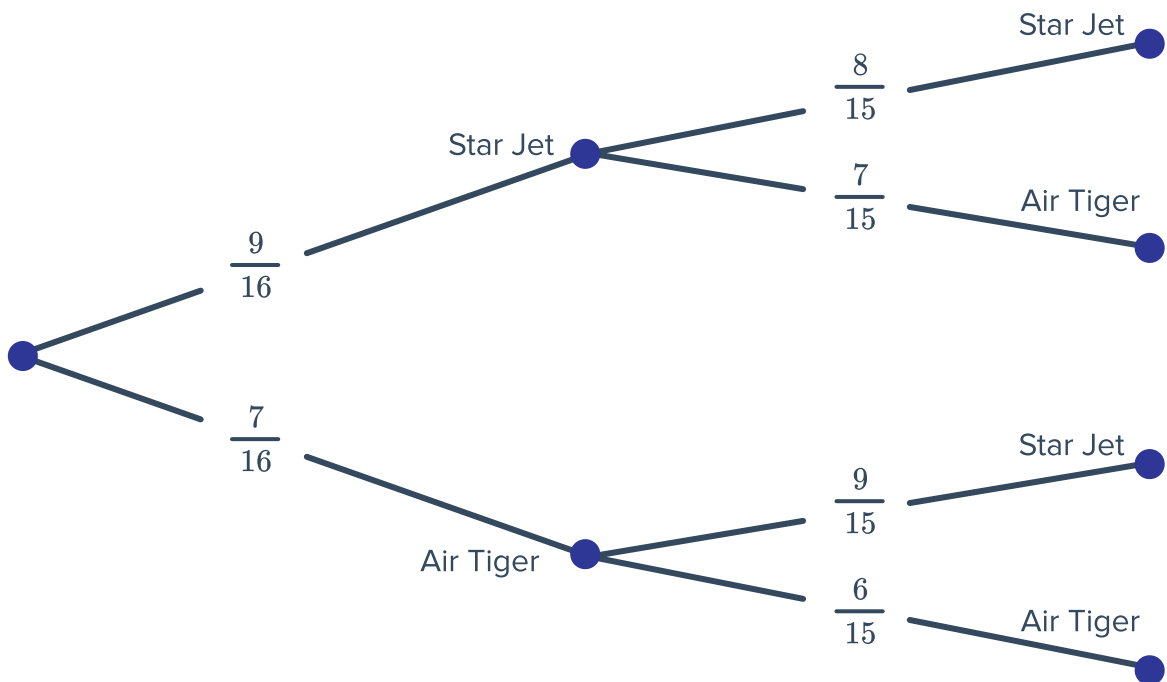
ii 0
vi $\frac{1}{4}$

iii $\frac{1}{2}$
vii $\frac{1}{4}$

iv $\frac{1}{3}$

25

a

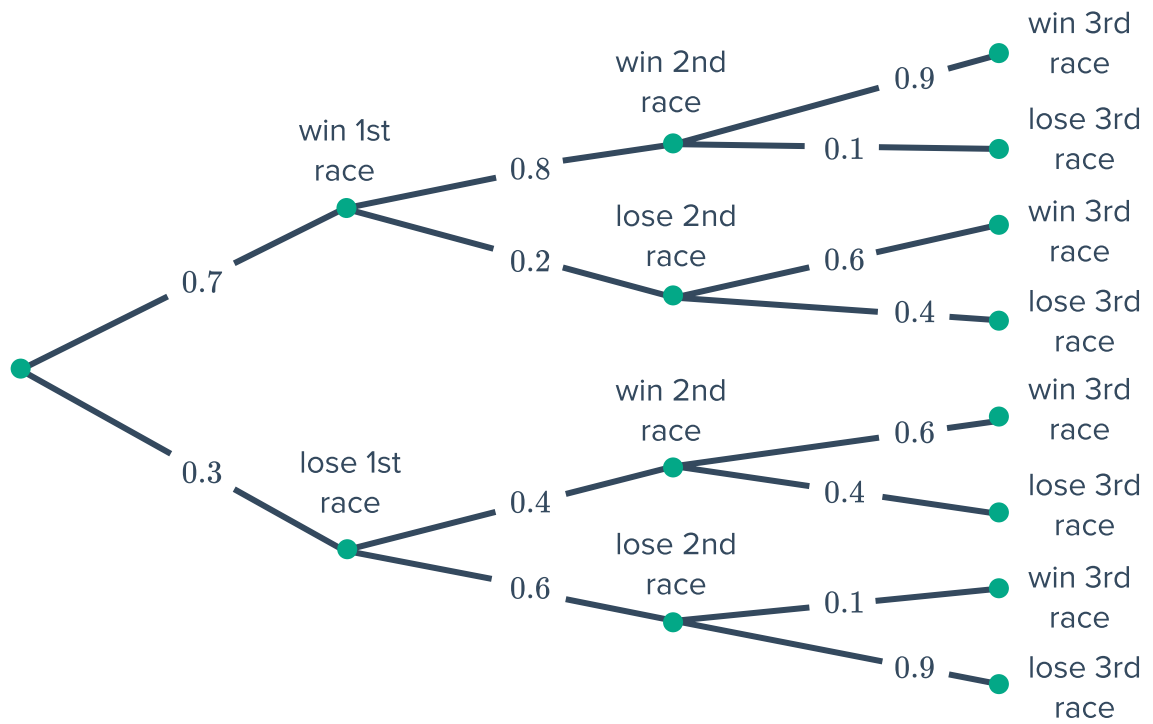


b

i $\frac{19}{40}$

ii $\frac{21}{40}$

26
a



b 0.504

c 0.678